



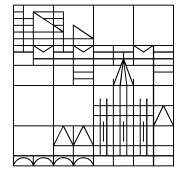
<https://github.com/pragmatic-test-kn/commitment-pragmatic-test>

SPP 2556: LaSTing

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Commitment through Accommodation: Evidence from LLMs

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Main Idea: Adapt Tiemann et al.'s (2015) study on the German presupposition trigger *wieder* ("again") to (L)LMs to test their sensitivity to context manipulations

Background

- Presuppositions are non-at-issue content, which can be associated with high levels of commitment (treated as entailments), especially when accommodated (?)
- Crucially, accommodation is known to be highly dependent on context
- We test whether (L)LMs distinguish between entailed and presupposed content of sentences with "again" when its presupposition is explicitly supported or not
- Previous studies on LMs show behaviour that diverges from human behaviour regarding presuppositions (Sieker & Zarriß, 2023)
- However, they seem to capture at least certain aspects of presuppositions
 - e.g., basic meaning and projection behaviour in NLI tasks (Parrish, et al. 2021; Sravanthi, et al., 2024; Jeretić, et al., 2020)

Items

- Context:** Last week Linda bought Judith a pink lamp for a room.
- Target:**
 - Yesterday, Judith received a pink lamp *again*. (verified)
 - Yesterday, Linda received a pink lamp *again*. (neutral)
- Interpretations (How often did J/L receive a pink lamp?):**
 - Once. (entailed)
 - At least twice. (presupposed)
 - It's impossible to tell.

Adaption to NLI task: context-target pairs serve as premises, interpretation options as separate conclusions

Sample NLI item (for verifying context):

P1: Last week, Linda bought Judith a pink lamp for a room.

P2: Yesterday, Judith received a pink lamp again.

C: Judith received a lamp once. → **entailed**

LLM Experiments

Methods:

I. Natural Language Inference:

- Models:** German/multilingual encoder models trained for NLI; multilingual instruction-tuned decoder models
- Task:** 1) + 2a) OR 2b) = premises, evaluate each conclusion 3) separately as either *entailment*, *contradiction* or *neutral*

II. Surprisal

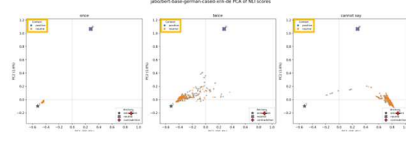
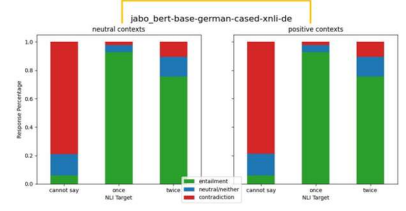
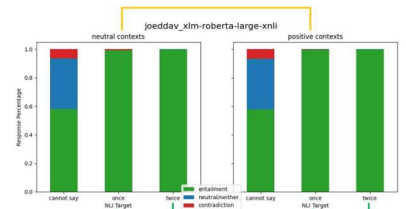
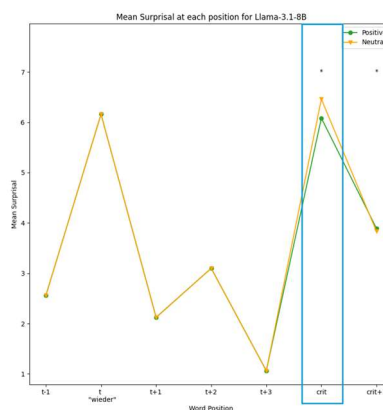
- Surprisal is said to predict human reading times (Smith and Levy, 2013)
- Models:** multilingual decoder models (base and instruction-tuned)
- Task:** given Premise 1, extract probability assigned to each word in Premise 2 by the model

$$4) \quad \text{Surprisal}(w) = -\log P(w|C)$$

where $P(w|C)$ is the probability of the word, given the preceding context

Results:

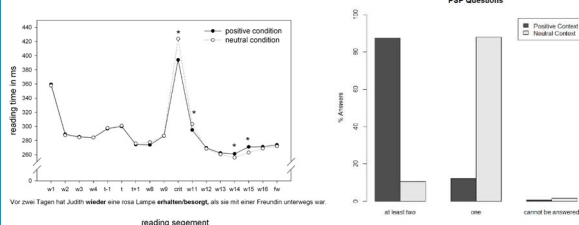
- NLI task
 - All models shows **no context distinction**
 - Most models show **no difference between the entailed and presupposed answers**
- Surprisal
 - Only Llama: significantly ($p < 0.01$) **higher surprisal** in neutral context on critical word
 - Higher surprisal on trigger and critical word



Human Experiments

Tasks and main findings (Tiemann et al., 2015):

- Acceptability rating of targets in given context
 - Result: neutral < verifying
 - Forced Choice between interpretative options in 3)
 - No sign of accommodation: once > twice in neutral context, twice > once in verifying context
 - Self paced reading time
 - On critical word: neutral < positive
- Suggests that people notice (and process) the presupposition, but they do not make choices that would indicate accommodation (cf. Tiemann, et al. 2011)



Summary and Discussion

- NLI task not parallel to human forced choice
 - LMs treat presupposed content as entailed; also in neutral conditions while humans seemingly ignore it
 - "cannot/impossible to say" option never used by humans, also inconclusive for LMs
- Surprisal ratings as a promising measure for comparison with human RTs also for presuppositions

Open Questions and Future Directions

- What tasks could show context effects and distinctions between different levels of (non-at-issue/at-issue) meaning?
- What are relevant model differences?
- Can we mimic cognitive effort seen in humans with other tasks?
- Do the NLI scatter plots correlate to different commitment levels in the LMs? How to test for levels of commitment with human subjects?

References: Tiemann, S., Kirsten, M., Beck, S., Hertrich, I., & Rolke, B. (2014). Presupposition processing and accommodation: An experiment on *wieder* ('again') and consequences for other triggers. Tiemann, S., Schmid, M., Bade, N., Rolke, B., Hertrich, I., Ackermann, H., Knapp, & Beck, S. (2011). Psycholinguistic evidence for presuppositions: On-line and off-line data. Smith, N. J., & Levy, R. (2013). The effect of word predictability on reading time is logarithmic. Parrish, A., Schuster, S., Warstadt, A., Agha, O., Lee, S. H., Zhao, Z., Bowman, S., & Linzen, T. (2021). NOPE: A corpus of naturally-occurring presuppositions in English. Sravanthi, S., Doshi, M., Tankala, P., Murthy, R., Dabre, R., & Bhattacharyya, P. (2024). PUB: A pragmatics understanding benchmark for assessing LLMs' pragmatics capabilities. Jeretić, P., Warstadt, A., Bhooshan, S., & Williams, A. (2020). Are natural language inference models IMPPRESSive? Learning IMPLICature and PRESupposition. Sieker, J., & Zarriß, S. (2023). When your language model cannot even do determiners right: Probing for anti-presuppositions and the maximize presupposition! principle. Tiemann, S. (2014). *The Processing of 'wieder' ('again') and Other Presupposition Triggers*.

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